

P2109R0: US084: Disallow "export import foo" outside of module interface

This paper resolves NB Comment US084:

`export import foo`; should not be allowed outside of module interface purview.

Both US033 and p1857 alter grammar and/or semantics of import declarations. The changes here presume both US033 and p1857 are accepted. p1857 introduced a problem with exported module-import-declarations. They ceased to be a thing. This wording also addresses that.

Wording:

Change [module.interface] para 1 (10.2)

export-declaration :
 export declaration
 export { declaration-seq_{opt} }
 export-keyword module-import-declaration

An *export-declaration* shall appear only at namespace scope and only in the purview of a module interface unit. An *export-declaration* shall not appear directly or indirectly within an unnamed namespace or a *private-module-fragment*. An *export-declaration* has the declarative effects of its *declaration*, ~~or its~~ *declaration-seq* (if any), or *module-import-declaration*. An *export-declaration* does not establish a scope and its *declaration* or *declaration-seq* shall not contain an *export-declaration* or *module-import-declaration*.

Para 2:

A declaration is *exported* if it is
— a namespace-scope declaration declared within an *export-declaration*, or
— a *module-import-declaration* ~~declared with the export keyword (10.3)~~ within an *export-declaration*, or
— a *namespace-definition* that contains an exported declaration, or
— a declaration within a header unit (10.3) that introduces at least one name.

Para 3:

An exported declaration that is not a *module-import-declaration* shall declare at least one name. If the declaration is not within a header unit, it shall not declare a name with internal linkage.

Change [module.import] para 1 (10.3), as already modified by us033 & p1857

module-import-declaration:
~~*export-keyword_{opt}*~~ *import-keyword module-name attribute-specifier-seq_{opt} ;*
~~*export-keyword_{opt}*~~ *import-keyword module-partition attribute-specifier-seq_{opt} ;*
~~*export-keyword_{opt}*~~ *import-keyword header-name attribute-specifier-seq_{opt} ;*

A *module-import-declaration* shall only appear at global namespace scope. In a module unit, all *module-import-declarations* and *export-declarations* exporting *module-import-declarations*, shall precede all other *declarations* in the *declaration-seq* of the *translation-unit* and of the *private-module-fragment* (if any). The optional *attribute-specifier-seq* appertains to the *module-import-declaration*.